Retriever-Conditioned Representation Search for Cross-Domain Patent-Family Retrieval

Anonymous manuscript

Abstract

Cross-domain prior-art retrieval must recover relevant patent families even when the query and target use different technical language. Representation is usually fixed before retriever evaluation, although field selection, segmentation, packing, and family aggregation determine what a retriever can score. We test whether representation should instead be selected conditional on a frozen retriever. Retriever-Conditioned Representation Search (RCRS) evaluates a finite grammar of executable patent-text programs without updating retriever weights. Five lexical or dense retrievers were screened on a family-level cross-domain benchmark. Development results varied by retriever: program transfer was asymmetric, one search surface collapsed to a single effective action, and fusing every primary system did not beat the strongest single system. One 125-query selection exposure fixed two systems for an 872-query confirmation. RCRS increased cross-domain (OUT) Recall@100 from 0.3311 to 0.4425 (paired difference 0.1114; 95% bootstrap confidence interval 0.1023–0.1204), with 619 wins, 158 ties, and 95 losses. The frozen selected RCRS configuration was then run on the complete 1,247-query benchmark. Of 5,193 strict OUT-relevant family pairs, 796 appeared in the first 100 ranks, 332 only in ranks 101–200, and 4,065 were absent from the Top-200 pool. Conditional representation search therefore produced a confirmed gain under the frozen protocol, while the post-confirmatory diagnosis placed most remaining strict OUT error outside the exposed candidate pool.

Keywords: patent retrieval, patent families, representation search, cross-domain retrieval, candidate exposure, reproducible evaluation

1. Introduction

A patent retrieval system rarely scores an invention as a single, uniform document. Technical content may be split across family members and across titles, abstracts, descriptions, and claims. Claims compress legal scope into specialized language; descriptions repeat concepts at several levels of detail. Cross-domain retrieval adds vocabulary shift between the query and the relevant family. These properties make text construction part of the retrieval configuration. Field order, structural labels, passage boundaries, packing, duplicate handling, and family aggregation all affect the evidence presented to a frozen retriever.

Family-level and embedding benchmarks now support controlled study of these choices. DAPFAM defines domain-aware relevance at patent-family level [1]. PatenTEB and a broader multi-task study compare patent embeddings across retrieval, classification, and clustering tasks [2, 3]. Those studies primarily compare models or embeddings. Here the model weights stay fixed while the executable representation changes.

AutoIndex frames representation construction as a program-search problem [4]. Patent records require a more constrained grammar than generic text: the program must respect document structure, long fields, family duplication, and candidate-level aggregation. RCRS combines that deterministic patent-family grammar with retriever-conditioned staged evaluation and a bounded confirmation of transfer behavior. Optimization also needs an evaluation protocol that separates iterative development from confirmation. Repeated access to one final set would make a representation program another route for test-set adaptation.

Retriever-Conditioned Representation Search (RCRS) addresses both points. The experimental repository records the implementation as ArmIndex; RCRS is the publication-facing name. The method searches serializable programs that transform patent text and aggregate family evidence. Retriever weights, training data, and relevance judgments do not change. The protocol separates iterative development, one Selection-125 exposure, paired Final-872 confirmation, and post-confirmatory analysis of the unchanged winner (Fig. 1).

Repository receipts use A1–A7 for study records; retriever identifiers remain in the provenance files. The article therefore uses step names and model names in the narrative and figures; Table 4 maps the study steps back to their repository records.

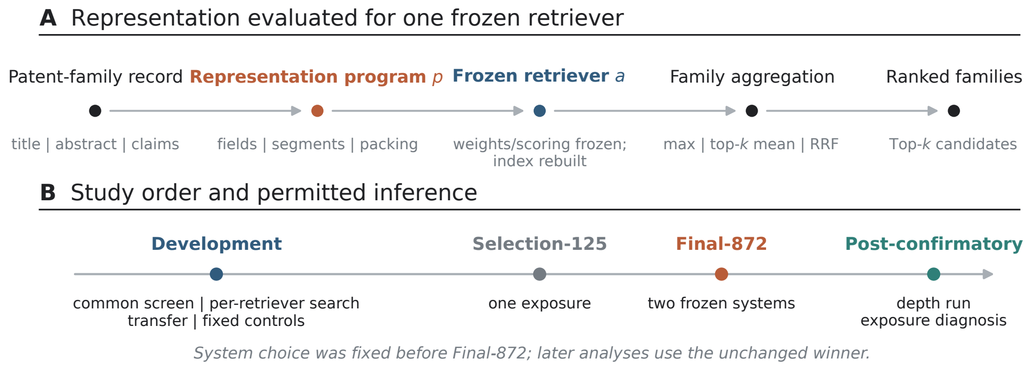

Figure 1: Method and study order. (A) An executable program constructs patent text and family-level evidence for one frozen retriever. Retriever weights and the scoring function are frozen; the derived candidate index is rebuilt deterministically for each representation program. (B) Development, Selection-125, Final-872 confirmation, and post-confirmatory analysis have separate inferential roles. System choice was fixed before Final-872.

The study answers three questions. First, does one representation program transfer across frozen retrievers, or are useful choices retriever-specific? Second, does the selected program improve a frozen system against a frozen static comparator on held-out confirmation? Third, after confirmation, how much strict OUT error reflects ordering inside the retrieved pool and how much reflects absent candidates? The claims remain within this family-level benchmark. We do not present a new embedding model, a component-level causal ablation, a universal baseline ranking, or a numerical comparison with external protocols that use different corpora or relevance definitions.

2. Related work

2.1. Patent-family retrieval and embedding evaluation

Patent retrieval differs from short-text web search in length, structure, duplication, and the cost of missed prior art. BM25 remains a useful lexical reference because it exposes exact-term evidence and has a well understood probabilistic basis [5]. Dense retrievers add semantic matching but impose model-specific context, pooling, and input conventions. Multilingual and patent-specialized embedding models therefore need evaluation at the same unit used by the application.

DAPFAM makes the family the evaluation unit and distinguishes cross-domain relations [1]. That design avoids counting several publications of the same invention as independent successes. PatenTEB and patent-embedding benchmarks cover a broader set of tasks and models [2, 3]. Their published scores cannot be placed in a common ranking with the present results unless corpus version, query construction, family normalization, relevance unit, and metric denominators match. They position the research question but do not serve as numerical comparators.

2.2. Executable representations, fusion, and candidate depth

An executable representation program records more than a prompt or a text template. It specifies the operations that produce indexable units and the rule that maps unit scores back to a patent family. AutoIndex provides the closest general formulation [4]. Our grammar specializes that idea to structured patent fields and ties every program to a frozen retriever.

Optimization in a compound system can produce flat or unstable search surfaces when distinct proposals compile to the same action or when a local change is incompatible with the surrounding components [6]. The evaluation therefore records the effective program signature and negative outcomes rather than counting proposals as independent progress. Rank fusion supplies a separate test of complementarity. Reciprocal rank fusion (RRF) combines lists without calibrated scores [7], but an extra list can still add noise. Best-single, top-two, top-three, all-primary, and commercial-only controls were fixed before evaluation.

Candidate depth separates first-stage exposure from later ordering. Passage retrieval can support fine-grained patent novelty analysis [8], but a downstream reranker or novelty model cannot act on an absent family. The post-confirmatory oracle therefore asks only what perfect ordering of the already frozen Top-200 could achieve. It is an analytical bound, not a reranker result.

3. Method

3.1. Retriever-conditioned objective

Let a denote a frozen retriever, $p \in \mathcal{P}$ an executable representation program, q a query patent record, and d a candidate patent-family record. Program p emits a finite set of candidate views,

$$\mathcal{V}_p(d) = \{v_1, \ldots, v_m\}. \tag{1}$$

The frozen retriever scores each view, and the program maps those scores to a family score,

$$S_{a,p}(q, d) = G_p(s_a(q, v_1), \ldots, s_a(q, v_m)) \, . \tag{2}$$

The development search selects

$$p_a^* = \arg\max_{p \in \mathcal{P}} J_a(p), \tag{3}$$

where J_a is the prespecified family-level objective for retriever a. OUT Recall@100 is primary; latency, cost, coverage, failures, and determinism are guardrails. The subscript a is substantive: a program selected for one retriever is not assumed to transfer to another.

For a query q, $R_q(q)$ denotes the query representation and $R_p(d)$ denotes the candidate-family views emitted by program p. Retriever weights and the scoring function are bound before evaluation. Changing p changes the candidate views and therefore the derived index; candidate indexes are rebuilt deterministically for each evaluated representation. The model, scoring function, relevance judgments, and metric code remain fixed.

For a query-level relevant-family set $\mathrm{Rel}(q)$, per-query recall and macro query-averaged recall are

$$r_q@k = \frac{|\, \mathrm{Rel}(q) \cap \mathrm{Top}_k(q)|}{|\, \mathrm{Rel}(q)|}, \qquad \mathrm{Recall}\,@k = \frac{1}{|Q|} \sum_{q \in Q} r_q@k. \tag{4}$$

The pair-level exposure diagnostic is different:

$$\mathrm{Exposure}\,@k = \frac{\sum_q |\, \mathrm{Rel}(q) \cap \mathrm{Top}_k(q)|}{\sum_q |\, \mathrm{Rel}(q)|}. \tag{5}$$

Macro query recall must therefore not be compared numerically with pair-level exposure.

3.2. Executable grammar and program identity

The grammar contains auditable text operations. A program selects title, abstract, and claims; orders fields; adds structural labels; creates document, section, claim, passage, or multiview units; sets passage size, overlap, and boundary handling; packs units within the retriever's input limit; normalizes Unicode, whitespace, case, and punctuation; applies a duplicate policy; and aggregates evidence by maximum score, top-three mean, top-two mean, or view-level RRF. The program cannot change retriever parameters, training data, the corpus, relevance labels, or metric code.

Every candidate program is serialized. Evaluation keys use its effective action signature, so syntactically different proposals that compile to the same behavior are counted once. This rule matters for the flat search surface reported in Section 5.

3.3. Finite representation-search procedure

The search used a frozen candidate manifest rather than an unconstrained live LLM sweep. The manifest contained 52 candidates: 40 matched candidates and 12 conditional reserves. Its proposal roles were exploit, matched-ablation, orthogonal, and diversity, and it was frozen before measurement with seed 20260812. Each frozen proposal was compiled, mapped to an effective action signature, and skipped if that signature had already been evaluated. For a new signature, the procedure constructed query and candidate views, evaluated the candidate on development data, and recorded valid, invalid, or dormant status with cost, coverage, and failures. The predeclared decision class (strict gain, numerical tie, or no strict gain) was then applied and the selected configuration was frozen. No proposal mutated a retriever, corpus, judgment, metric, or later split.

The A2 accounting was 52 frozen candidates, 44 valid measured candidates, 8 dormant reserves, and zero declared scientific failures. The separate A3 HarnessOpt run used three complete batches and 12 frozen proposals; all proposals collapsed to one effective executable action signature, so the search surface stopped at that flat boundary. These counts describe distinct development records and are not additive.

Table 1: Aggregate-safe accounting for finite representation search. The candidate manifest was frozen before measurement; dormant reserves were not measured.

Record	N	Unique sigs.	Valid	Dorm.	Decision
A2 per-retriever search	52	not exposed	44	8	strict gain; tie; no strict gain
A3 HarnessOpt	12	1	–	–	flat-surface stop

3.4. Frozen retrievers

All retrievers were frozen before their reported evaluations. They span a lexical baseline, a general multilingual embedding model, a patent-specialized model, and two recent general embedding models (Table 2). Context limits are recorded configuration properties; they do not imply equal effective coverage of a patent record. PatEmbed uses CC BY-NC-SA 4.0, which limits commercial deployment. The license is therefore part of the system boundary, not a footnote to the effectiveness result.

Table 2: Frozen retrievers. Short names are used in the text and figures; repository retriever identifiers remain in the provenance files.

Short name	Retriever	Type; dimension; context	License	Role in development
BM25	BM25S 0.3.10	lexical; –; –	–	diagnostic lexical baseline [5]
BGE-M3	BAAI/bge-m3	dense multilingual; 1,024; 8,192	MIT	diagnostic [9]
PatEmbed	datalyes/ patembed-large	patent dense; 1,024; 512	CC BY-NC-SA 4.0	advanced; research/non-commercial
Arctic	Snowflake Arctic-Embed-M-v2.0	dense multilingual; 768; 8,192	Apache 2.0	advanced [10]
Qwen3	Qwen3-Embedding-0.6B	dense multilingual; 1,024; 32,768 model-declared tokens	Apache 2.0	advanced [11]

Table 3: The two systems compared on Final-872. Human-readable field, segmentation, pooling, packing, and index settings are not included in the safe projection; the hashes bind the frozen configurations without exposing protected payload.

System	Human-readable binding	Frozen configuration identifiers
Selected RCRS	research champion; PatEmbed; research-only license	winner binding `285e94b1...`; representation `bcbffffa...`; retrieval `e2ba107f...`
Static comparator	static common baseline; exact two-system comparator	system `9a98a6d8...`; program `00f1c295...`; prompt `51a81f1b...`

The ellipsized identifiers in Table 3 are display prefixes of canonical SHA-256 values; the complete values remain in the owner-controlled receipts. The table does not imply unreported field or pooling settings. Final-872 is therefore an exact comparison of these two frozen systems, not a claim against an exhaustive baseline suite.

4. Experimental design

4.1. Benchmark and evidence stages

Every experiment belongs to the same family-level cross-domain benchmark lineage [1]. The benchmark construction is publication/source records → benchmark family records → ordered text fields → normalization and duplicate policy → views/passages → one family score. The complete retrieval pool contains 45,336 candidate families. Table 4 records population, exposure, freeze state, and permitted interpretation for each study step. Development is iterative. Selection-125 is the only selection exposure. Final-872 compares exactly two frozen systems. The depth run and candidate-exposure diagnosis use only the confirmed winner after Final-872.

4.2. Patent-family construction and split protection

The benchmark treats a candidate family as the unit of retrieval and evaluation. Publication/source records are grouped into benchmark-defined family records; ordered text fields are extracted, normalized, and subjected to the frozen duplicate policy before views or passages are formed. View scores are then aggregated to one family score. The safe publication projection does not expose member identifiers, language/member-selection rules, or near-duplicate overlap details, so those implementation settings remain an owner gate rather than being inferred here.

The frozen split ledger is summarized below. Development comprises the REP-DEV 150-query cohort, the HDEV-100 100-query cohort, and their combined Train-250 250-query development record. Selection-125 contains 125 queries, with OUT metrics on the verified (n=90) eligible subset. Final-872 contains 872 paired queries. Split membership, judgments, and hashes were frozen before the corresponding exposure and remain protected. The safe package does not expose query-family overlap identifiers or a zero-overlap audit; it therefore does not claim zero cross-stage family overlap.

4.3. Development, transfer, and fixed controls

The common screen applies one representation to all five retrievers. The per-retriever search then preserves each recorded decision class (strict improvement, tie, or no strict improvement). The transfer study applies each

Table 4: Study steps, repository records, populations, and claim boundaries. Repository codes identify experimental records, not retrievers.

Record	Study step	Population	System state and exposure	Permitted inference
A1–A3	Development: common screen, per-retriever search, transfer and controls	development cohorts	five retrievers; iterative search; fixed transfer and fusion controls	describe retriever interaction, transfer, complementarity, ties, and negative results
A4	Selection-125	125 queries; OUT primary $n = 90$	at most four frozen profiles; one exposure	select one profile
A5	Final-872 confirmation	872 queries	two frozen systems; no tuning; paired by query	estimate the confirmatory effect of RCRS against the static comparator
A6	Full-benchmark depth run	1,247 queries	confirmed winner only; depth 200; after Final-872	report depth-wise stress test; no comparator effect
A7	Candidate-exposure diagnosis	strict relation-scoped IN/OUT; OUT has 905 judged queries and 5,193 positive pairs	unchanged winner and its frozen Top-200 pool	diagnose candidate exposure and the within-pool ordering bound

Table 5: Aggregate split and exposure ledger. OUT counts are metric-population counts, not total stage membership when otherwise stated.

Split	Queries	Development record	OUT population	Exposure	Prot.
Development	150 + 100; 250 combined	REP-DEV; HDEV; Train-250	development-specific	iterative	frozen aggregates
Selection-125	125	–	OUT-eligible (n=90)	one exposure	protected
Final-872	872	–	OUT-eligible; 17,429 pairs	held out after freeze	protected

advanced-retriever program to every advanced target retriever without editing. The 3×3 matrix tests compatibility; it is not a second selection procedure.

Five fixed controls test whether rank-list complementarity changes the result: the strongest single system, top-two RRF, top-three RRF, all-primary fusion, and a commercial-only profile. The all-primary and top-three controls resolve to the same aggregate scores in the canonical artifact. None of these development controls can replace Selection-125 or enter Final-872 after inspection.

4.4. Selection and confirmation

Selection-125 evaluates the frozen research reference, BALANCED, DEEP, and FAST profiles on 125 queries. The primary OUT subset contains 90 queries. One exposure chooses the research profile. That profile and one static-representation comparator are then frozen for Final-872. Confirmation requires 872/872 coverage and paired query-level comparison; a failure would remain visible in the report.

4.5. Metrics, uncertainty, and population accounting

Recall@k is computed at patent-family level. OUT Recall@100 is primary. nDCG@100 is a confirmatory secondary outcome. nDCG@10 is descriptive because the canonical Final-872 artifact contains no confidence interval for that endpoint. nDCG uses the benchmark's binary relevance grades. Final-872 differences are paired by query. Percentile 95% confidence intervals use 10,000 paired bootstrap resamples of the paired query units; win, tie, and loss counts use the same query unit. Development decision classes are strict gain, numerical tie, and no strict gain, as recorded in the aggregate ledger.

The notation distinguishes two denominators. OUT_{eligible} is the eligible-out Final-872 cohort and contains 17,429 relevant pairs. OUT_{strict} is the relation-scoped post-confirmatory population with 905 judged queries and 5,193 positive pairs. IN and OUT can overlap at query level because one query may have both relation types. Final-872 and the full-benchmark depth run therefore describe different populations; they are not a before–after series.

5. Development results

5.1. Common screen and per-retriever search

Common-screen Recall@100 was 0.1912 for BM25, 0.2699 for BGE-M3, 0.4134 for PatEmbed, 0.3407 for Arctic, and 0.3637 for Qwen3. The corresponding nDCG@100 values were 0.1727, 0.2314, 0.3478, 0.2845, and 0.3079. BM25 and BGE-M3 remained diagnostic; PatEmbed, Arctic, and Qwen3 entered advanced development. Appendix Table A.7 gives the full common-screen and per-retriever aggregate ledger.

The per-retriever search produced different outcomes. Arctic improved from 0.3407 to 0.3587 Recall@100 and met its strict-improvement rule. Qwen3 rose from 0.3637 to 0.3737 but did not meet that rule. PatEmbed reached 0.4230 and retained the recorded classification "numerical tie to common screen." BM25 and BGE-M3 also tied under their decision rules, at 0.2347 and 0.2900. Program utility therefore depended on the frozen retriever.

5.2. Transfer, fusion, and effective search surface

Figure 2A reports all nine advanced-retriever transfers. The three programs produced a narrow Recall@100 range on PatEmbed (0.4184–0.4193), a lower range on Arctic (0.3374–0.3413), and 0.3595–0.3626 on Qwen3. No program source held a consistent diagonal advantage, and the target-retriever ordering remained visible after transfer.

The diagonal cells in Fig. 2A are descriptive transfer cells, not the per-retriever winner values in Appendix Table A.7. The transfer matrix and the ledger use different development cohort and configuration summaries: the matrix applies each source program unchanged to a target retriever, whereas the ledger reports the separate per-retriever search record. Their diagonal values therefore need not agree, and no value is corrected or promoted from one record to the other.

The controls were also non-monotonic (Fig. 2B). Best-single and top-two RRF were nearly tied in Recall@100 (0.4184 and 0.4187), while top-two RRF increased nDCG@100 from 0.3471 to 0.3527. All-primary fusion and top-three RRF both returned 0.4151 Recall@100 and 0.2848 nDCG@10, below the strongest single system on those endpoints. The commercial-only profile returned 0.3693 Recall@100. It quantifies a licensing trade-off within this benchmark; it is not an alternative Final winner.

The HarnessOpt procedure proposed 12 candidates in three batches, but their effective action signatures collapsed to one. Additional proposal rounds did not create additional executable behavior. Nominal proposal count would therefore overstate the amount of search; the effective signature is the reported unit.

6. Selection and confirmatory results

6.1. *One selection exposure*

All four Selection-125 profiles completed 125/125 queries. The research reference obtained 0.4161 OUT Recall@100. BALANCED and DEEP tied at 0.3606; FAST obtained 0.3083. Against either BALANCED or DEEP, the paired difference was 0.0556 (95% CI 0.0300–0.0817; 49 wins, 20 ties, 21 losses). Against FAST, the difference was 0.1078 (95% CI 0.0811–0.1350; 69/9/12). The canonical Selection-125 artifact contains no verified nDCG result, so none is inferred. The selected RCRS configuration and static comparator were frozen after this exposure.

6.2. *Final-872 effectiveness*

Both Final-872 systems completed 872/872 queries without failure. RCRS obtained 0.4425 OUT Recall@100; the static system obtained 0.3311 (Fig. 3). The paired difference was 0.1114 (95% CI 0.1023–0.1204), with 619 wins, 158 ties, and 95 losses. nDCG@100 increased from 0.2793 to 0.3656, a paired difference of 0.0863 (95% CI 0.0787–0.0941). Descriptive nDCG@10 increased from 0.2337 to 0.2975, an aggregate difference of 0.0638.

Under the frozen Final-872 protocol, conditional representation search improved the selected system over its static comparator. The estimate applies to this paired comparison; external retrievers and individual grammar components were outside its design.

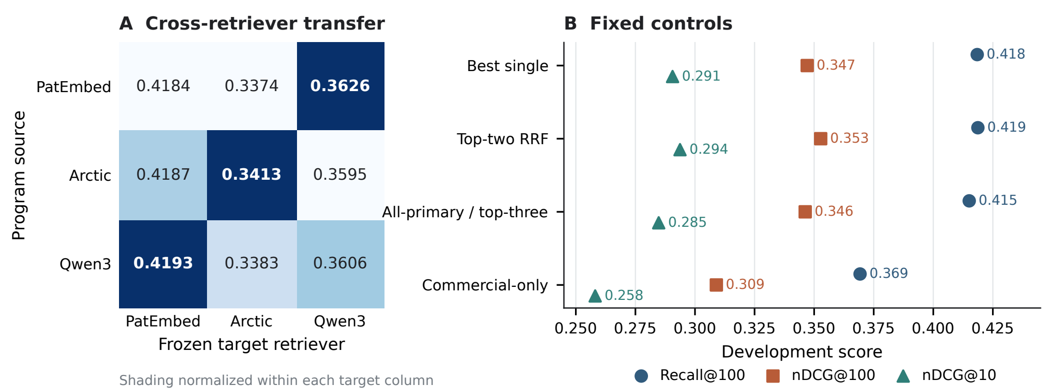

Figure 2: Development-only transfer and fixed controls. (A) Recall@100 for programs optimized on PatEmbed, Arctic, or Qwen3 and applied to each frozen target retriever; shading is normalized within a target column. (B) Fixed fusion and licensing controls. These data were not used to revise the Final-872 systems.

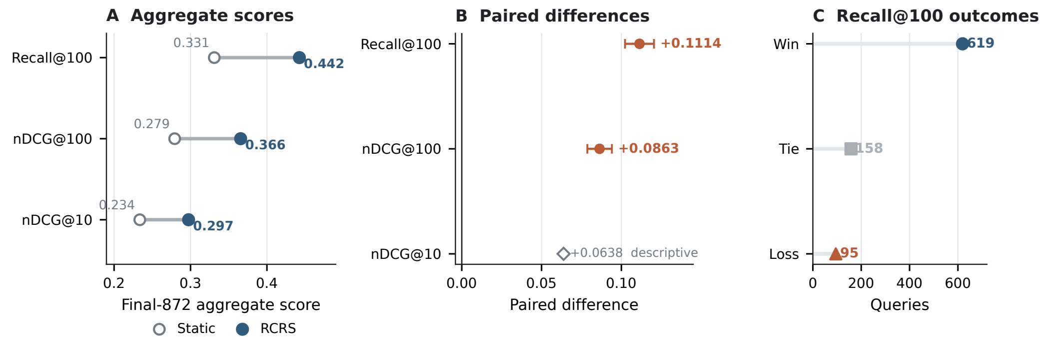

Figure 3: Final-872 comparison of two frozen systems. (A) Aggregate effectiveness. (B) Paired query-level differences and 95% bootstrap confidence intervals; nDCG@10 is descriptive. (C) Query-level Recall@100 wins, ties, and losses.

Table 6: Final-872 effectiveness and operational checks. Confidence intervals are paired query-level bootstrap intervals. The p99 latency result is retained although it favors the comparator.

Outcome	Static	RCRS	Difference or ratio	Statistical or operational note
OUT Recall@100	0.3311	0.4425	+0.1114	CI 0.1023–0.1204; 619/158/95
OUT nDCG@100	0.2793	0.3656	+0.0863	95% CI 0.0787–0.0941
OUT nDCG@10	0.2337	0.2975	+0.0638	descriptive; no canonical interval
Latency p50 / p95 (ms)	315.862 / 735.171	237.547 / 732.218	–	lower median; similar p95
Latency p99 (ms)	804.002	1,044.613	–	higher for RCRS
Throughput	2.665918 q/s	3.319344 q/s	1.245	direct throughput ratio (research/static)
Equal-cost resource metric	–	–	1.456818	ratio computed at the same canonical cost (USD 1.456818 each); not a throughput ratio
Coverage / failures	872/872; 0	872/872; 0	–	both deterministic in the recorded run

The operational results were mixed. RCRS reduced median latency and had a similar p95, but its p99 was 1,044.613 ms versus 804.002 ms for the static system. Throughput was higher for RCRS (3.319344 versus 2.665918 queries per second), a direct ratio of 1.245. The separate equal-cost resource metric is 1.456818 because both systems have the same canonical cost estimate of USD 1.456818; it is not the throughput ratio. These are benchmark-scale measurements, not evidence of patent-office deployment scalability. The worse tail latency prevents an unqualified efficiency claim.

7. Post-confirmatory diagnosis

7.1. Complete-benchmark depth profile

After Final-872 fixed the selected RCRS configuration, the full-benchmark depth run evaluated that system alone on 1,247 queries and 45,336 candidate families to depth 200. The run produced 249,400 ranked rows with no failure. Overall Recall@10, @20, @50, @100, and @200 was 0.1368, 0.2147, 0.3364, 0.4390, and 0.5468 (Fig. 4A). IN recall at those depths was 0.1870, 0.2779, 0.4131, 0.5282, and 0.6451. Strict OUT recall was 0.0340, 0.0709, 0.1340, 0.1884, and 0.2602. nDCG@100 was 0.3625 overall, 0.4065 for IN, and 0.0706 for OUT_{strict}.

The depth run contains no comparator and uses relation-scoped denominators. It describes the frozen selected RCRS configuration at increasing candidate depth; it does not re-estimate the Final-872 effect.

7.2. Strict OUT exposure and Top-200 bound

The candidate-exposure diagnosis partitions 5,193 OUT_{strict}-relevant family pairs (Fig. 4B). Of these, 796 appeared in ranks 1–100, 332 only in ranks 101–200, and 4,065 were absent at rank 200. Among 905 judged OUT_{strict} queries, 67 were fully retrieved by rank 200, 297 were partially retrieved, 86 had relevant evidence only in ranks 101–200, and 455 had no relevant candidate in the Top-200 pool. Appendix Table A.8 records these counts.

The analytical oracle moves each relevant family already present in the frozen Top-200 into the first 100 positions; it adds no candidate. The resulting Recall@100 bounds were 0.5468 overall, 0.6451 for IN, and 0.2602 for OUT_{strict}, compared with observed values of 0.4390, 0.5282, and 0.1884. The within-pool gaps were 0.1079, 0.1169, and 0.0717 (Fig. 4C). These are ordering bounds, not achieved reranker scores. Even perfect ordering of the exposed Top-200 would leave 4,065 strict OUT_{strict}-relevant pairs unavailable.

8. Discussion

8.1. Representation is part of the retriever configuration

The development results reject a model-independent account of preprocessing. Arctic met its strict-improvement rule; PatEmbed and Qwen3 did not. Transfer changed scores by target retriever and produced no consistent source-program advantage. These outcomes fit a conditional view: the useful text units and aggregation rule depend on the scoring behavior and input constraints of the retriever that consumes them.

Final-872 shows that this interaction can survive a freeze. The estimate is paired, its interval excludes zero, both systems cover the same 872 queries, and the comparison was fixed after one selection exposure. The negative and tied development outcomes are part of the same result. They show that executable search can find a better configuration without guaranteeing that every retriever, every proposal batch, or every fusion will improve.

8.2. Candidate exposure is a separate engineering target

The exposure decomposition narrows the next question. A reranker can reorder the 332 strict OUT-relevant pairs found only in ranks 101–200 and can change the placement of relevant pairs already in the first 100. It cannot recover the 4,065 pairs absent from the Top-200. Candidate generation, coverage, and cross-domain exposure therefore need separate measurement from ordering quality.

The oracle is reported as a bound. Calling the Top-200 bound a reranking result would imply an implemented model and an attainable gain. Neither exists here. The bound measures only the largest Recall@100 value permitted by the frozen candidate pool.

8.3. Implications for patent-search evaluation

Three reporting practices follow from the evidence. A representation program should be versioned with its retriever, not listed as generic preprocessing. Family normalization, duplicate handling, passage packing, and score aggregation should appear in the executable configuration. The benchmark's citation-derived relevance relations are evaluation labels, not legal findings about novelty or inventive step. Finally, relation-scoped populations should be reported with their query and pair denominators. The same label, such as OUT, is insufficient when protocols use different eligibility rules.

Licensing also constrains system choice. PatEmbed supports the research comparison but its non-commercial share-alike terms may exclude a production deployment. The commercial-only control quantifies one in-benchmark alternative, but it does not establish the best deployable system. A production study would need a new prespecified comparison among models with compatible terms.

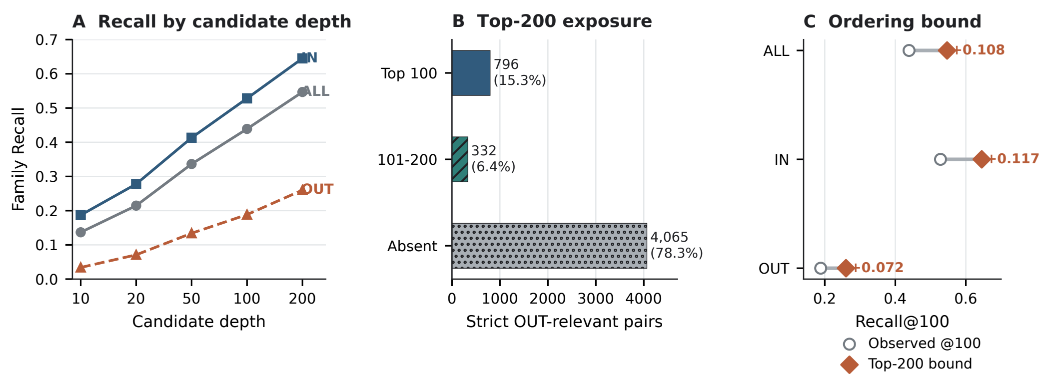

Figure 4: Diagnosis of the unchanged selected RCRS configuration after confirmation. (A) Family-level recall by candidate depth. (B) Pair-level exposure of 5,193 OUT_{strict}-relevant pairs. (C) Observed Recall@100 and the analytical bound from perfect ordering of the frozen Top-200 pool. IN and OUT are relation-scoped and may overlap by query.

8.4. Limitations and follow-up work

The evidence is bounded in six ways. It comes from one family-level benchmark with 45,336 candidate families, not a patent-office-scale deployment. The benchmark-scale latency and throughput measurements should not be read as production capacity evidence. Citation-derived relevance is not a legal novelty or inventive-step determination. Final-872 contains one frozen comparator, not an exhaustive baseline suite. The grammar changes several components jointly, and no hash-bound component ablation identifies a causal field, segmentation rule, or aggregation operation. External benchmark configurations were not verified as comparable, so cross-study scores are not ranked. IN and OUT are relation-scoped and can overlap by query. The candidate-exposure diagnosis contains no implemented rescue model or layer-three reranker.

A separate preregistered study could test grammar ablations, additional licensed baselines, an independent benchmark, and candidate-generation methods with a new development/final split. Reusing Final-872 for those decisions would weaken the confirmation reported here.

9. Conclusion

Selecting an executable patent-text program for a frozen retriever increased OUT_{eligible} Recall@100 by 0.1114 over a frozen static representation on Final-872. The gain is specific to the tested family-level protocol. After confirmation, the unchanged selected RCRS configuration exposed only 1,128 of 5,193 OUT_{strict}-relevant pairs in its Top-200, placing most of the remaining error outside the pool available to a reranker. The method result and the exposure limit are separate: RCRS improves representation under a freeze, while broader candidate coverage remains an open retrieval problem.

Declaration of competing interest

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Author roles will be inserted after author approval and before submission.

Data and code availability

The benchmark remains subject to its source terms, and PatEmbed remains subject to CC BY-NC-SA 4.0. The submission package contains aggregate tables, deterministic figure-generation code, frozen configuration identifiers, and provenance paths for every reported number. It does not redistribute query identifiers, patent or family identifiers, relevance judgments, per-query outcomes, or raw rankings. The blinded repository URL, archive identifier, and final availability statement will be added after owner review.

Declaration of generative AI and AI-assisted technologies

During manuscript preparation, the authors used an AI-assisted drafting tool for language and document-structure support. The authors reviewed and edited the output, checked reported results against canonical artifacts, and retain full responsibility for the article. This statement will be reconciled with the journal policy in force on the submission date.

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Appendix A. Aggregate evidence ledger

The appendix is part of the main manuscript. No supplementary scientific file is required. Tables A.7 and A.8 contain aggregate values only.

Table A.7: Common screen (repository A1) and per-retriever search (repository A2). Decision classes reproduce the canonical artifact and are not inferred from displayed decimal changes.

	Common screen			Per-retriever search			
Retriever	R@100	nDCG@100	nDCG@10	R@100	nDCG@100	nDCG@10	Decision class
BM25	0.191200	0.172717	0.160011	0.234667	–	–	diagnostic tie
BGE-M3	0.269933	0.231377	0.198497	0.290000	–	–	diagnostic tie
PatEmbed	0.413400	0.347812	0.289856	0.423000	0.357636	0.299444	numerical tie
Arctic	0.340667	0.284546	0.235538	0.358667	0.301868	0.253229	strict gain
Qwen3	0.363733	0.307930	0.256706	0.373667	0.321262	0.273664	no strict gain

Table A.8: Candidate-exposure diagnosis (repository A7). The strict OUT pair-level total is 5,193; the judged-query total is 905.

Pair-level class	Count	Query-level class	Count
Relevant pair in Top-100	796	fully retrieved by rank 200	67
Relevant pair in ranks 101–200	332	partially retrieved by rank 200	297
Relevant pair absent at rank 200	4,065	deep-only at ranks 101–200	86
		no relevant candidate in Top-200	455